

COLUMBUS RICKENBACKER, OH, USA

WMO: 724285

Lat: 39.817N

Lon: 82.933W

Elev: 744

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 13812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.7	(c) 10.1	(d) -3.7	(e) 4.6	(f) 7.9	(g) 1.4	(h) 6.1	(i) 11.7	(j) 27.0	(k) 36.1	(l) 24.8	(m) 34.7	(n) 6.0	(o) 270	(p) 0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.3	(c) 92.2	(d) 73.9	(e) 90.2	(f) 73.3	(g) 88.0	(h) 72.4	(i) 79.6	(j) 86.1	(k) 77.5	(l) 85.6	(m) 75.6	(n) 82.5	(o) 9.5	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.2	150.7	83.9	75.1	135.4	80.9	73.2	126.8	79.1	43.9	86.2	41.3	85.3	39.7	81.0	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.8	(b) 20.0	(c) 17.7	(e) -2.4	(f) 96.1	(g) 7.7	(h) 6.4	(i) -8.0	(j) 100.6	(k) -12.5	(l) 104.4	(m) -16.9	(n) 107.9	(o) -22.5	(p) 112.6		
(4) 23.8	(5) 20.0	(6) 17.7	(7) -2.4	(8) 96.1	(9) 7.7	(10) 6.4	(11) -8.0	(12) 100.6	(13) -12.5	(14) 104.4	(15) -16.9	(16) 107.9	(17) -22.5	(18) 112.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	30.4	33.6	42.6	54.3	64.5	72.8	75.7	74.4	68.0	55.8	43.9	35.1
	DBStd	18.38	12.41	11.19	11.00	9.05	8.22	5.96	4.78	4.89	6.84	8.82	9.29	10.10
	HDD50	2174	617	470	287	57	3	0	0	0	0	39	228	473
	HDD65	5030	1072	880	698	336	114	10	0	2	47	310	633	928
	CDD50	3770	9	10	57	186	454	685	797	757	540	218	46	11
	CDD65	1152	0	0	4	15	100	246	332	294	137	24	0	0
	CDH74	10314	0	1	34	159	926	2184	3009	2564	1213	221	3	0
	CDH80	3731	0	0	5	25	291	819	1182	946	413	50	0	0
(14) Wind	WSAvg	7.3	8.9	8.6	8.8	8.8	7.3	6.6	5.5	5.1	5.6	6.9	7.7	8.1
(15) Precipitation	PrecAvg	40.4	2.9	2.2	3.4	3.9	4.2	4.1	4.3	3.4	3.3	2.8	2.6	3.1
	PrecMax	53.0	8.5	4.6	7.6	7.4	7.9	9.1	10.4	7.7	6.1	6.3	5.0	7.2
	PrecMin	28.8	1.1	1.0	1.7	1.3	1.1	0.9	1.8	1.0	1.1	0.8	0.7	1.3
	PrecStd	5.7	1.5	0.9	1.3	1.3	1.9	1.8	1.7	1.7	1.4	1.3	1.1	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	67.4	79.0	84.5	90.2	93.2	95.8	94.9	93.1	86.0	72.5	64.5
		MCWB	59.9	58.7	62.7	65.0	68.9	73.7	75.9	74.4	70.3	68.8	60.5	59.8
	2%	DB	59.1	60.9	71.9	79.3	86.7	90.5	91.6	91.0	88.4	80.6	67.8	59.2
		MCWB	55.4	53.5	59.3	62.4	68.7	73.0	75.5	73.5	71.1	66.6	58.2	55.0
	5%	DB	54.2	55.2	65.5	74.7	83.5	88.2	90.0	88.3	84.5	75.2	63.4	54.7
		MCWB	51.2	50.6	56.2	60.4	68.3	72.2	74.5	73.1	69.6	63.8	56.8	51.1
	10%	DB	47.7	49.9	60.7	70.4	79.5	84.8	87.2	85.5	81.4	70.9	59.2	50.2
		MCWB	44.5	44.7	52.9	58.4	66.3	70.3	73.3	71.5	68.0	62.0	52.9	46.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.6	60.8	64.4	71.0	76.7	80.5	82.9	82.0	77.5	72.4	64.8	61.1
		MCDB	63.4	66.6	73.6	78.8	83.8	87.2	89.8	88.4	84.8	81.0	69.4	64.0
	2%	WB	56.5	55.5	60.8	66.1	72.6	77.5	79.9	79.0	74.7	69.2	60.8	55.5
		MCDB	59.1	57.8	69.5	72.8	80.2	86.3	86.8	85.2	82.4	75.8	65.4	58.6
	5%	WB	51.3	50.9	57.9	63.0	70.2	75.3	77.7	76.8	72.6	66.6	57.5	51.9
		MCDB	53.5	55.2	64.0	69.5	78.4	82.8	85.8	84.1	80.3	72.7	61.9	54.4
	10%	WB	45.0	45.4	54.0	60.2	68.2	73.4	75.7	74.9	70.7	63.6	53.8	46.5
		MCDB	47.3	48.7	59.6	68.1	76.9	80.6	82.4	81.0	77.5	69.0	58.7	49.0
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	15.9	18.2	21.4	20.4	19.5	19.3	19.9	21.3	20.2	17.4	13.6
		MCDBR	18.7	23.5	26.1	26.4	24.6	23.1	22.1	23.3	25.9	25.6	23.7	18.8
		MCWBR	15.9	18.2	16.1	14.5	10.9	9.4	9.0	9.2	10.4	13.4	16.1	15.9
	5% WB	MCDBR	17.7	22.2	22.0	21.9	21.1	19.6	19.5	18.9	20.5	20.6	19.9	18.4
		MCWBR	15.8	18.9	15.5	14.8	12.1	10.0	10.2	10.0	10.1	12.4	15.7	16.9
(40) Clear-Sky Solar Irradiance	taub		0.312	0.331	0.360	0.397	0.441	0.458	0.467	0.460	0.400	0.366	0.329	0.307
	taud		2.422	2.344	2.336	2.258	2.169	2.166	2.152	2.152	2.299	2.386	2.454	2.456
	Ebn at Noon		270	280	283	279	267	262	258	257	266	264	262	263
	Edh at Noon		26	32	36	42	47	47	47	46	37	31	25	23
(44) All-Sky Solar Radiation	RadAvg		526	790	1113	1504	1739	1907	1851	1721	1407	977	664	448
	RadStd		50	103	100	131	144	129	104	93	129	89	87	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-4.90	N/A	-2.97	-3.96	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+232	+128

Nomenclature: See separate page